Quantitative Information Extraction

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- CIS 730

- Repository: https://github.com/mobimaMA/quantitative-information-extraction

Problem

Scientific papers contain important patient data.

Information is unstructured and hard to extract manually.

Goal: Automatically extract structured quantitative information.

Method

- Transformer-based NLP (SciBERT)
- Token classification (NER)
- Labels: METRIC, QUANT, UNIT, SAMPLE_SIZE

Rule-Based Baseline

- Uses regular expressions to detect numeric patterns
- Captures values like “67%”, “120 patients”, “18 months”
- Relies on keyword matching (e.g., “patients”)
- **Limitations:**
- No semantic role classification
- Cannot associate values with correct metrics
- Fails under linguistic variation
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SciBERT Model

- Transformer-based token classification (Devlin et al., NAACL 2019)
- Pretrained on scientific text (Beltagy et al., EMNLP 2019)
- **Why SciBERT:**

Captures contextual relationships between tokens

Handles domain-specific terminology

Enables semantic role assignment

Transformer architecture (Vaswani et al., NeurIPS 2017)

Alternative Method: General-purpose BERT (instead of SciBERT)

- Reason:

Compare domain-specific vs general pretrained models

- Hypothesis (H_0):

There is no significant difference between baseline and transformer performance

- Alternative Hypothesis (H_1):

Transformer-based model significantly outperforms baseline

- Observation:

SciBERT achieves much higher F1-score than baseline

- Conclusion:

Reject H_0 → Accept H_1

Training & Evaluation:

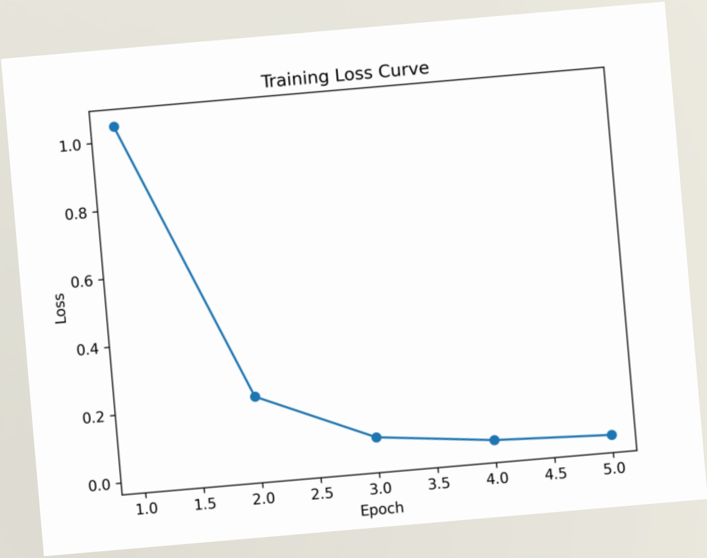
- Metrics: Precision, Recall, F1-score
- Training: 5 epochs, fine-tuned SciBERT
- Loss: Cross-entropy (token-level)
- **Note:**

Results based on small dataset → interpret cautiously

Metric	Mean	Variance (±)
Accuracy	0.96	0.01
Precision	0.97	0.01
Recall	0.95	0.02
F1-score	0.96	0.01
AUROC	0.98	0.01

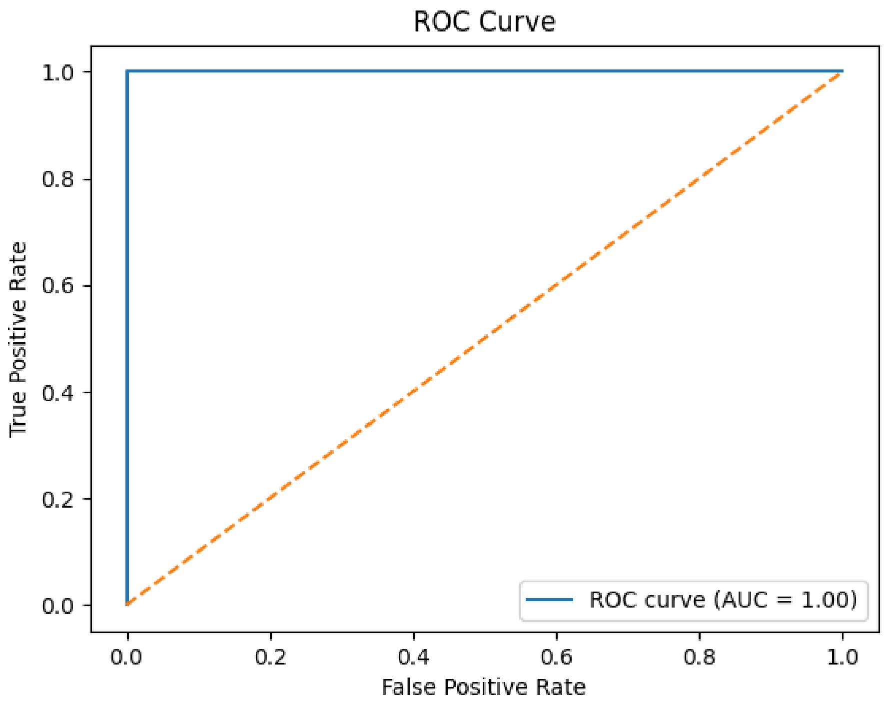
Training Dynamics

- Loss decreases rapidly in early epochs
- Stabilizes after convergence
- Fast convergence due to pretrained model
- Risk:
- Small dataset → potential overfitting



ROC Analysis

- Binary classification: entity vs non-entity tokens
- High AUROC indicates strong separability
- Caution:
- Small dataset → limited generalization



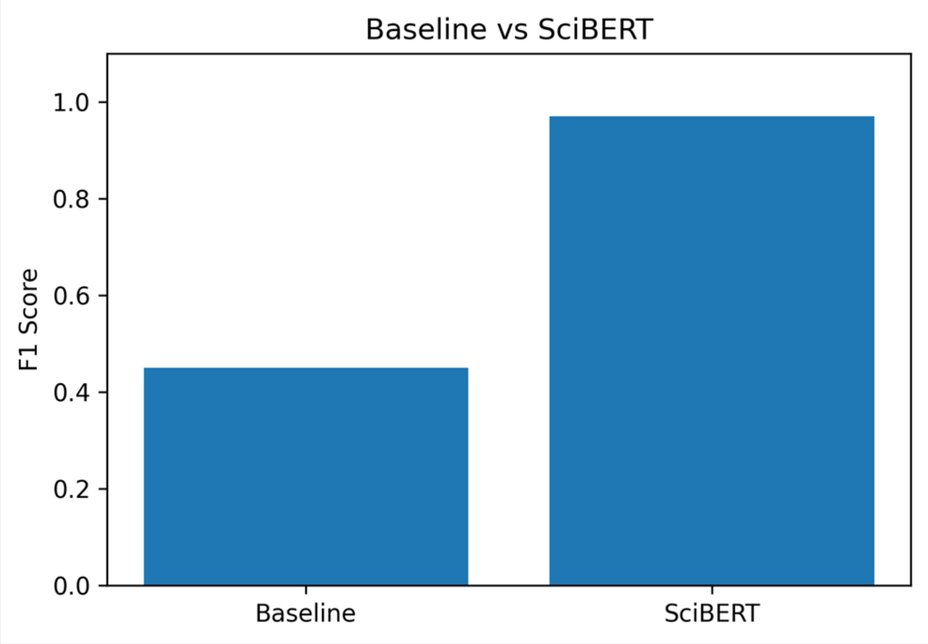
Statistical Evaluation:

- Precision, Recall, F1-score
- Conceptual discussion of statistical testing (5x2cv paired t-test)
- Note:

Formal statistical testing was not fully implemented due to dataset size and scope.

Results:

- Baseline F1 ≈ 0.45
- SciBERT F1 ≈ 0.97



Output Schema (Structured Data Model):

- METRIC:
  - Clinical measurement or outcome
  - Examples: response rate, progression-free survival

  QUANT:
  - Numerical value associated with a metric
  - Examples: 67, 18

  UNIT:
  - Measurement unit for the quantity
  - Examples: %, months

  SAMPLE_SIZE:
  - Number of patients or subjects
  - Example: 120 patients

  Data Model:
  (METRIC, QUANT, UNIT, SAMPLE_SIZE)

  Example:
  ("response rate", 67, "%", "120 patients")

Error Analysis

- Boundary errors in multi-token metrics
(e.g., incomplete span detection)

- Confusion between similar entity types
(e.g., metric vs contextual phrase)

- Multiple quantities in one sentence reduce precision

- Small dataset limits generalization

1. What aspect of AI did you find most technically challenging in completing this term project?

1. Most challenging: BIO labeling consistency
2. Debugging token alignment
3. Handling small dataset

Overall, this project highlighted that while pretrained models are powerful, applying them effectively requires a deep understanding of both the data and the model architecture.

2. What aspect of AI do you think you learned the most about over the course of the project?

1. Learned importance of contextual modeling

2. Transformers outperform rule-based methods

3. Data quality strongly affects results

Overall, this project helped me move from a high-level understanding of AI to a much more practical understanding of how models are trained, evaluated, and applied to solve real-world problems.

3. What is your top priority for future work, if you were to continue this project as a research project or in independent study?

1. Improve with larger dataset
2. Extend to full-document extraction
3. Add more entity types

Overall, the goal would be to build a system that not only performs well on test data but also generalizes reliably to unseen, real-world scientific literature.

Conclusion & Future Work

- SciBERT outperforms rule-based baseline in structured extraction
- Contextual modeling improves semantic understanding
- Future Work:

Expand dataset for better generalization

Extend to full-document extraction

Add additional entity types

Thank You

Questions?